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Computer Graphics (CSIT 401)

III semester, BScCSIT

Compiled by **Ankit Bhattarai**
Ambition Guru College

Syllabus

Unit	Contents	Hours	Remarks
1.	Introduction to Computer Graphics	5	
2.	Scan Conversion Algorithms	7	<i>Very Important</i>
3.	2D Transformations and Viewing	6	<i>Very Important</i>
4.	3D Transformations and Viewing	5	
5.	3D Object Representation	6	
6.	Solid Modeling	3	
7.	Visible Surface Detection	5	
8.	Illumination, Shading and Rendering Techniques	5	
9.	OpenGL and Graphics Programming	3	

Practical Works

Credit hours : 3

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Laboratory Works

1. DDA Line Algorithm
2. Bresenham line drawing algorithm
3. Mid Point Circle Algorithm
4. Mid Point Ellipse Algorithm
5. Basic transformation on 2D including Translation, Rotation and Scaling
6. Simple 3D Object with basic transformations including Translation, Rotation and Scaling
7. Clipping
8. Hidden surface removal
9. Basic Drawing Techniques in OpenGL

Resources

Resources

1. Some chapters in pdf form.
2. Question Collection & Assignments (50-80 Questions)

Text Books/Reference Books:

- Donald Hearn & Pauline Baker, Computer Graphics with OpenGL, 3rd/4th Edition, Pearson Education, 2011.
- Edward Angel, Interactive Computer Graphics: A Top-Down Approach with OpenGL, 5th Edition, Pearson Education, 2008.
- Tomas Akenine-Möller et al., Real-Time Rendering, A K Peters/CRC Press.
- John F. Hughes et al., Computer Graphics: Principles and Practice.

Let's
visualize
computer
graphics

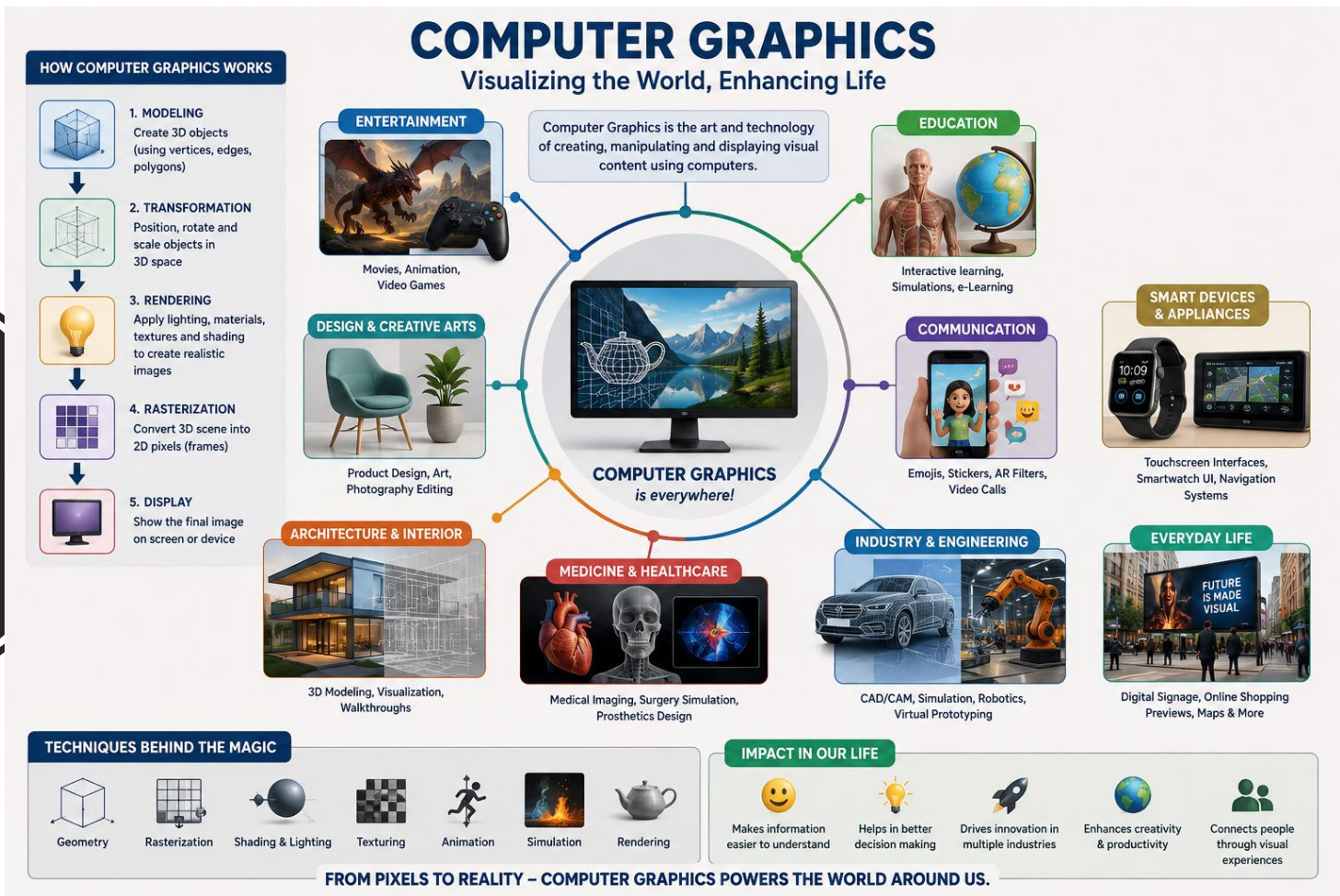


Figure 1: Computer Graphics and Its Applications in Everyday Life
Source: Image generated with ChatGPT (OpenAI) 2026 based on the author's prompt.

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Unit 1

Introduction to Computer Graphics (5 hrs.)

Introduction and Applications of Computer Graphics, Evolution, Overview of Graphics Hardware(Monitors, GPUs, modern Display Technologies: LCD, OLED, VR/AR Systems, GPU and Graphics Accelerators), Raster and Random Scan Displays and Architecture, Display Processor, Graphics Pipeline: From Modeling to Rendering, Brief History: From Raster/CRT to Real-Time Rendering, Input Devices, Graphics Software, Need for Machine-Independent Graphics Language

Computer Graphics

Definition:

- Computer graphics is a field of study and practice that involves the creation, manipulation, and representation of visual images and animations using computers.

Computer Graphics = Data structure + Graphics Algorithm + Language

- Data Structure means those data structure that are suitable for computer graphics.
- Graphics algorithm refers to algorithm for picture generation and transformation.
- Language means high level language for generation of graphics or patterns of objects.

Evolution of Computer Graphics

Computer Graphics: History of Computer Graphics

1950s–1960s: Early Foundations

- Birth of computer graphics through experiments with digital computers.
- **1950: Ben Laposky** created the first electronic images using an oscilloscope.
- **1963: Ivan Sutherland** developed **Sketchpad**, the first interactive computer graphics system.
- Introduced concepts such as graphical user interaction, object manipulation, and CAD foundations.

1960s–1970s: Graphics Hardware Development

- Emergence of specialized graphics hardware and display systems.
- Development of **frame buffers**, **graphics terminals**, and **CRT displays**.
- **1972: Ed Catmull** created *A Computer Animated Hand*, one of the first 3D computer animations.
- Computer graphics began to be used in scientific visualization and engineering.

Computer Graphics: History of Computer Graphics

1980s: Rise of 3D Graphics

- Introduction of powerful graphics workstations from companies such as Silicon Graphics (SGI).
- Significant advances in rendering, shading, and geometric modeling.
- **1982:** *Star Trek II: The Wrath of Khan* showcased advanced CGI effects.
- Computer graphics became important in movies, simulations, and design.

1990s: CGI Revolution

- Rapid growth of 3D modeling and animation software.
- **1995:** Pixar released **Toy Story**, the first fully computer-animated feature film.
- Real-time rendering technologies improved dramatically.
- Widespread adoption of computer graphics in film, television, and advertising.

Computer Graphics: History of Computer Graphics

Gaming Era (1970s–Present)

- **1972:** Release of **Pong**, one of the first commercial video games.
- **1978:** **Space Invaders** popularized arcade gaming.
- **1993:** **Doom** introduced immersive 3D environments.
- **1996:** **Quake** advanced real-time 3D graphics and multiplayer gaming.
- Modern games feature realistic graphics, physics, and virtual worlds.

2000s–Present: Modern Graphics

- Development of programmable GPUs and advanced graphics APIs.
- Real-time ray tracing, AI-assisted rendering, and photorealistic graphics.
- Growth of **Virtual Reality (VR)** and **Augmented Reality (AR)**.
- Applications expanded to healthcare, education, engineering, entertainment, and mobile devices.

Computer Graphics Vs Image Processing

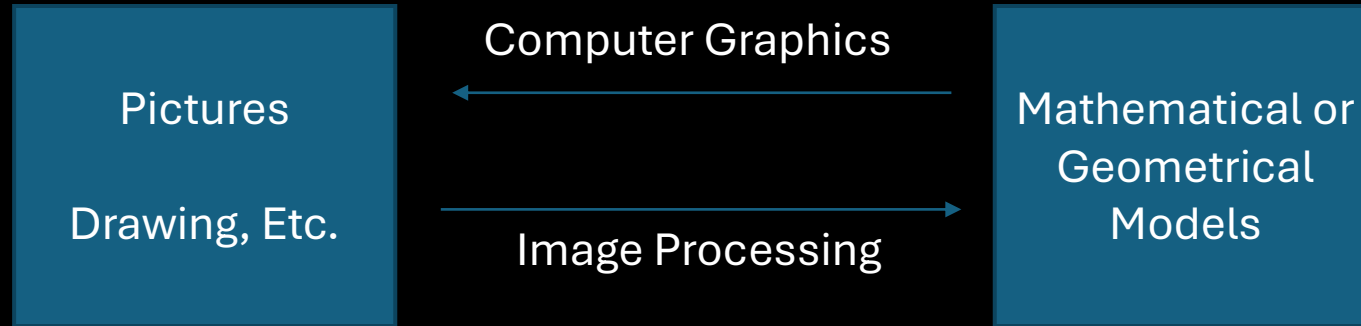
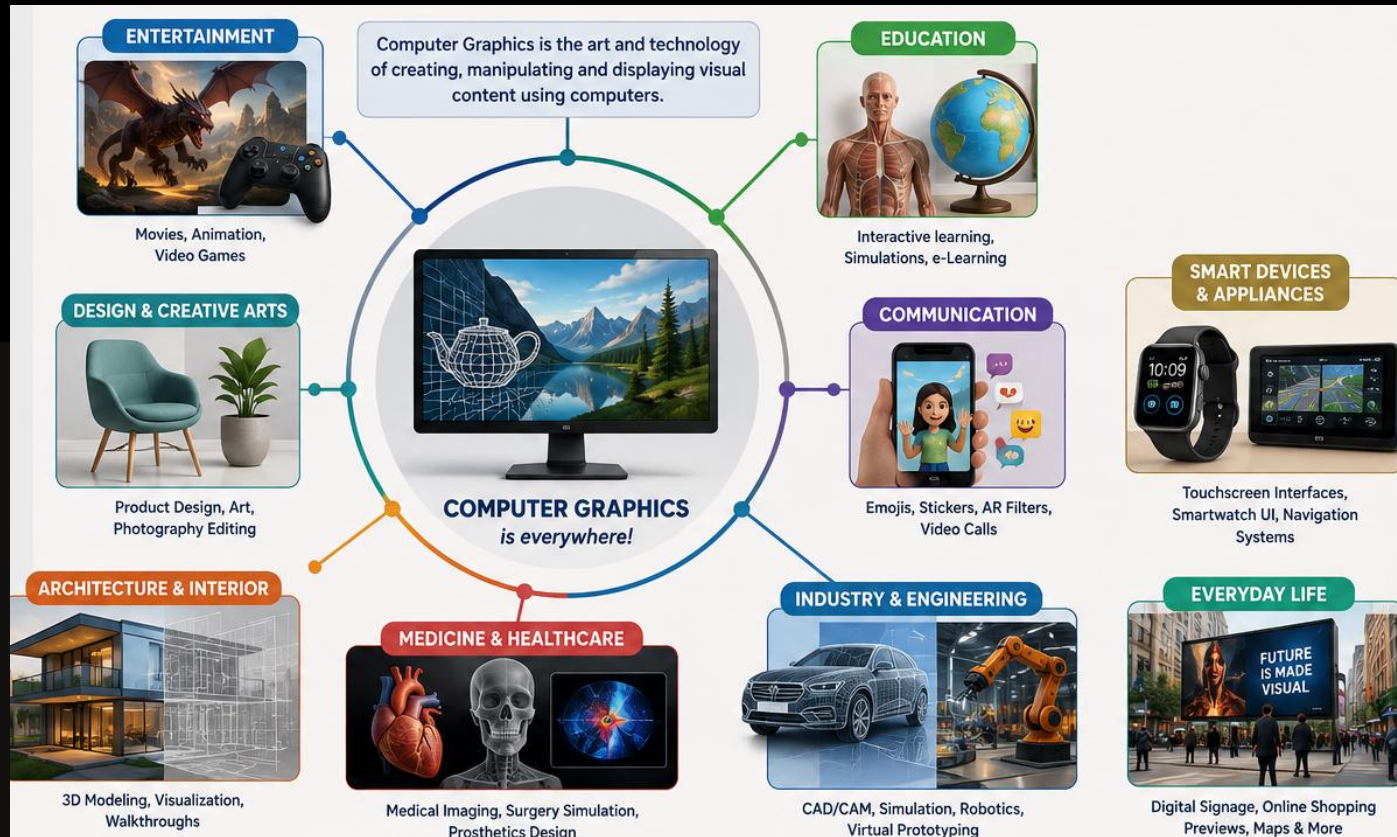


Figure 2. Computer Graphics and image processing

Computer Graphics Vs Image Processing

Computer Graphics	Image Processing
Field related to generation of images or pictures using computers	Applies techniques to manipulate the existing pictures
Synthesizes pictures from the mathematical or geometrical models.	It analyzes the existing pictures to derive description in mathematical or geometrical forms.
It includes the creation parts of images.	It includes the manipulation of images.
E.g. Drawing any kind of picture	E.g. Making the picture blur to use it for specific task.

Applications of computer graphics



Applications of computer graphics

Computer Graphics : Applications of computer graphics

- **Entertainment and Media:**

- Video Games:** Computer graphics are fundamental to the gaming industry, creating realistic 2D and 3D environments, characters, and special effects.
- Movies and Animation:** CGI (Computer-Generated Imagery) is used extensively in film and animation for creating visual effects, animated characters, and entire virtual worlds.

- **Design and Visualization:**

- Computer-Aided Design (CAD):** Architects, engineers, and designers use computer graphics to create and visualize complex structures, products, and systems.
- Interior and Exterior Design:** Designers use 3D graphics to model and visualize interior and exterior spaces, helping clients to better understand design concepts
- Fashion Design:** Fashion designers use digital tools to create and render clothing designs before production.

Computer Graphics : Applications of computer graphics

- **Scientific Visualization:**

- **Medical Imaging:** Computer graphics play a crucial role in visualizing medical data, such as MRI and CT scans, to aid in diagnosis and surgery planning.
- **Astronomy:** Astronomers use computer graphics to visualize celestial objects and phenomena, enabling better understanding and analysis.
- **Geographical Information Systems (GIS):** GIS software uses graphics to map and analyze geographic data for various applications, including urban planning and environmental monitoring.

- **Education and Training:**

- **Simulators:** Flight simulators, medical simulators, and military training tools use computer graphics to create realistic training environments.
- **Educational Software:** Graphics enhance educational materials, making learning more engaging and interactive.

Computer Graphics : Applications of computer graphics

- **Virtual Reality (VR) and Augmented Reality (AR):**

- **VR:** Computer graphics are the foundation of immersive VR experiences, allowing users to interact with and explore virtual environments.
- **AR:** AR overlays computer-generated graphics onto the real world, offering applications in gaming, navigation, and information display.

- **Advertising and Marketing:**

- **Graphic Design:** Graphics are used in advertisements, branding, and marketing materials to convey messages and attract attention.
- **Product Visualization:** Companies use computer graphics to showcase products in 3D, allowing customers to interact with them online before making purchases.

Computer Graphics : Applications of computer graphics

- **Simulation and Modeling:**

- **Weather Forecasting:** Meteorologists use computer graphics to simulate and visualize weather patterns for forecasting and analysis.
- **Fluid Dynamics:** Engineers and scientists use graphics to model and understand fluid behavior in applications like aerodynamics and hydrodynamics.

- **Art and Digital Media:**

- **Digital Art:** Artists create digital paintings, sculptures, and multimedia installations using computer graphics software.
- **Graphic Design:** Graphic designers use computer graphics for creating logos, posters, websites, and other visual content.

Computer Graphics : Applications of computer graphics

- **Gaming and Interactive Experiences:**

- **Interactive Exhibits:** Museums and exhibitions use computer graphics to create interactive displays and educational experiences.
- **Theme Parks:** Attractions often employ computer graphics to create immersive rides and shows.

- **Communication and Data Visualization:**

- **Data Visualization:** Graphics help represent complex data sets in an understandable and visually appealing manner.
- **User Interfaces (UI):** UI design relies on graphics to create user-friendly and visually pleasing software interfaces.

Computer Graphics : Applications of computer graphics

- **Cartography**

- Cartography is a subject which deals with the making of maps and charts. Computer graphics is used to produce both accurate and schematic representations of geographical and other natural phenomena from measurement data. Examples include geographic maps, oceanographic charts, weather maps, contour maps and population density maps
- Surfer is one of such graphics packages which is extensively used for cartography



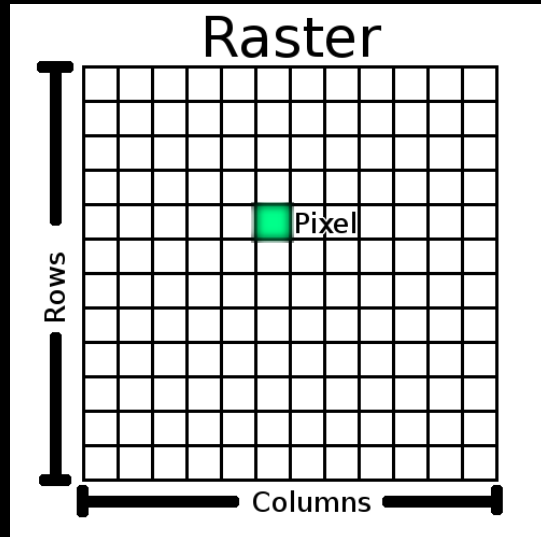
Computer Graphics: Pros and Cons

Advantages	Disadvantages
Enhancing visual communication	Requiring specialized skills and software
Facilitating the creation of complex designs	Potentially leading to unrealistic or exaggerated representations
Improving the realism of visual content	Taking time to create
Enabling the creation of 3D models and animations	Potentially promoting a reliance on technology
Improving the accessibility of visual content	Being vulnerable to manipulation

Overview of Graphics Hardware(Monitors, GPUs, modern Display Technologies: LCD, OLED, VR/AR Systems, GPU and Graphics Accelerators)

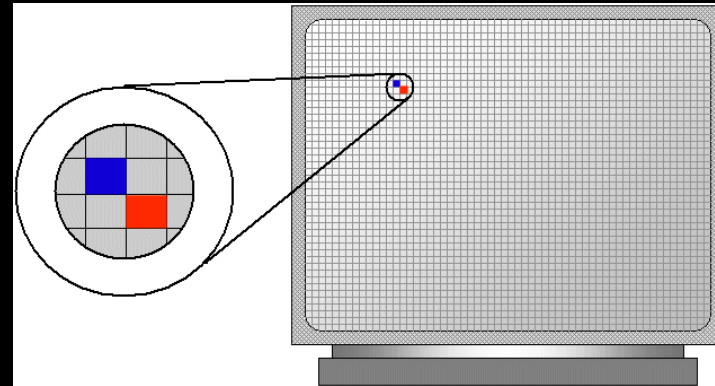
Some terminologies

Raster: A rectangular array of points or dots.

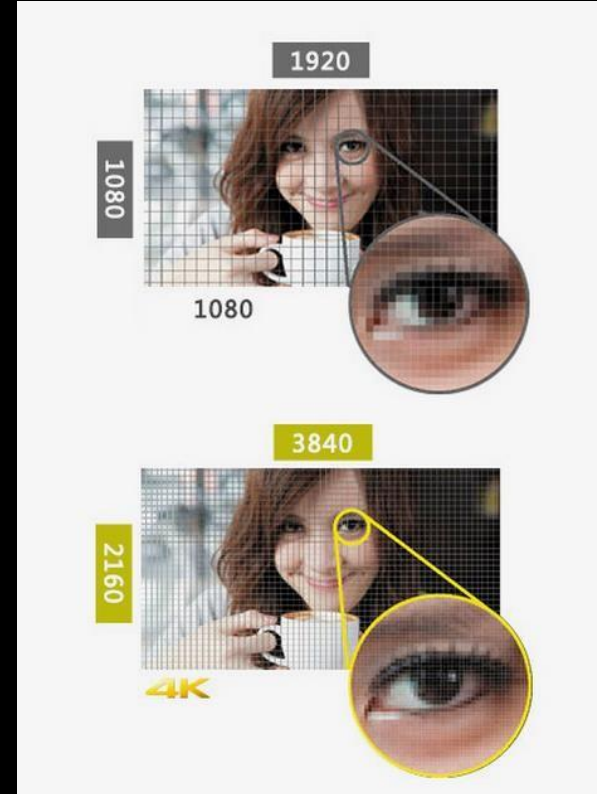
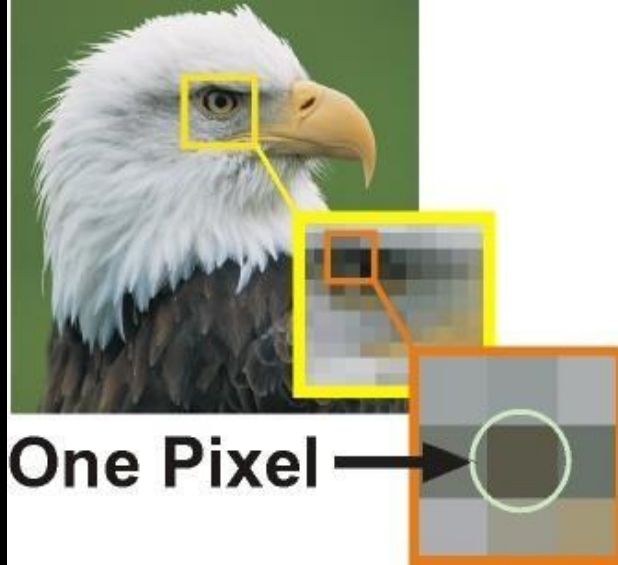


Pixel:

- One dot or picture element of the raster
- The pixel (a word invented from "picture element") is the basic unit of programmable color on a computer display or in a computer image



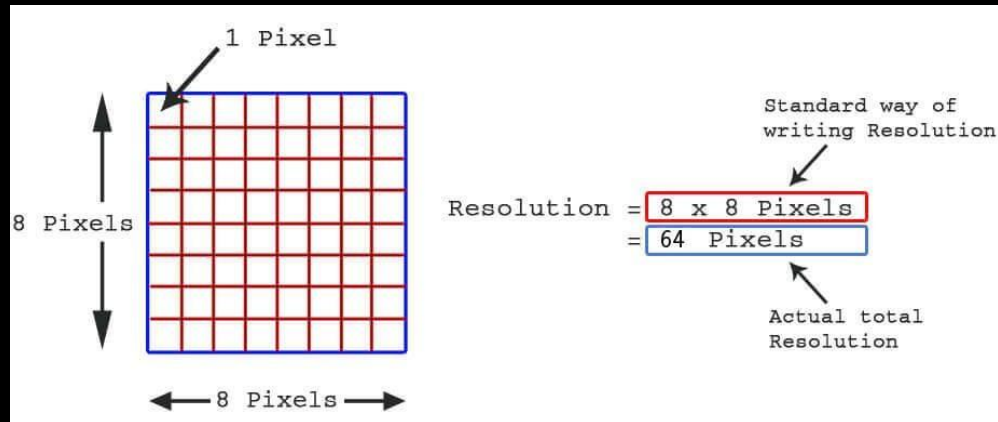
Pixel



Resolution

- The maximum number of points (pixel) that can be displayed without overlap on a CRT is referred to as the resolution.
- It is also defined as the **number of points per unit of measure (per centimeter or per inch) that can be plotted** horizontally and vertically.
- Resolution is defined as the maximum member of points that can be displayed horizontally and vertically without overlap on a display device.

Resolution



Resolution



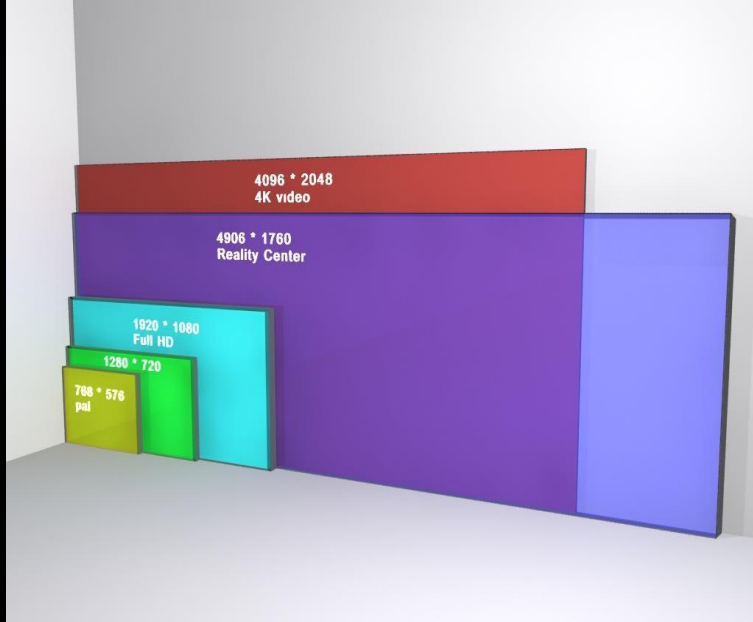
Here's a quick overview of the most common resolutions:

- SD: 480p / 576p
- HD: 720p / 1080p
- 4K: 2160p
- 8K: 4320p

Full Aperture 4k	4096 x 3112	12,746,752 pixels
Academy 4k	3656 x 2664	9,739,584 pixels
Digital Cinema 4k	4096 x 1714	7,020,544 pixels
Digital Cinema Aperture 4k	3996 x 2160	8,631,360 pixels

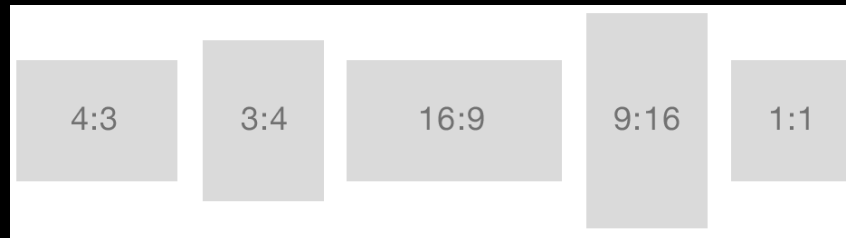
Resolution

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Aspect Ratio

- The **aspect ratio** of an image describes the proportional relationship between its width and its height.
- It is commonly expressed as two numbers separated by a colon, as in $16:9$. For an $x:y$ aspect ratio, no matter how big or small the image is, if the width is divided into x units of equal length and the height is measured using this same length unit, the height will be measured to be y units.
- In, for example, a group of images that all have an aspect ratio of $16:9$, one image might be 16 inches wide and 9 inches high, another 16 centimeters wide and 9 centimeters high, and a third might be 8 yards wide and 4.5 yards high.



Refresh Rate

- The number of times the screen is redrawn each second.
- Higher refresh rates mean less flicker on the screen, which translates into less eyestrain.



Numerical

Q.N.1: If the pixel values are accessed from the frame buffer with an average access time (for one single pixel) of 20 ns and the total resolution of the screen is 1024 X 800 , will there be a flickering effect seen on the screen ?



time (for one
s 1024 X 800

To glow all pixel on screen it takes (T)

	= 1024 X 800 X 20 ns
	= 16,384,000 ns
	= 0.01638 second

$$\begin{aligned}\text{frequency}(F) &= 1 / T \\ &= 1 / 0.01638 \\ &= 61.05 \text{ Hz}\end{aligned}$$

since it is above 50 times/sec ***there will be no flickering effect seen on screen.***

Q.N.2: In case of raster system with resolution 1024 X 1280, how many pixel could be accessed per second in the system by a display controller at a rate of 60 frames per second? What is accessed time per pixel in this system ?

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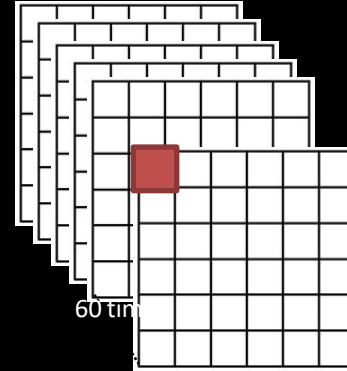
Ans :

1. Pixel accessed :

= 1024 X 1280 X 60 pixel can be
accessed in this system

2. Access time per pixel:

= $1/(1024 \times 1280 \times 60)$
= 12.71ns



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Q.N.3.: How long would it takes to load a 640 X 480 frame buffer with 12 bit per pixels if 10^5 bits can be transferred per second.

***Formula:**

Required size of frame= Resolution or total no. of pixel $\times$ no. of bit per pixel on screen

To calculate the time it takes to load a frame buffer, you can use the formula:

$$\text{Time} = \frac{\text{Number of bits in the frame buffer}}{\text{Transfer rate}}$$

First, calculate the number of bits in the frame buffer:

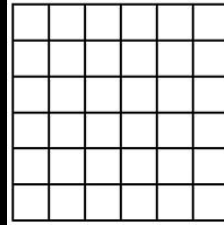
$$\text{Number of bits} = \text{Width} \times \text{Height} \times \text{Bits per pixel}$$

Q.N.3.: How long would it takes to load a 640 X 480 frame buffer with 12 bit per pixels if 10^5 bits can be transferred per second.

Ans :

total size of frame buffer :
= $640 \times 480 \times 12$

it takes (Time in sec) = $(640 \times 480 \times 12) / 10^5$



***Formula:**

Required size of frame= Resolution or total no. of pixel $\times$ no. of bit per pixel on screen

To calculate the time it takes to load a frame buffer, you can use the formula:

$$\text{Time} = \frac{\text{Number of bits in the frame buffer}}{\text{Transfer rate}}$$

First, calculate the number of bits in the frame buffer:

$$\text{Number of bits} = \text{Width} \times \text{Height} \times \text{Bits per pixel}$$

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Q.N.4: Consider three different raster system with resolution 640 by 400, 1280 by 1024 and 2560 by 2048. what size frame buffer (in byte) is needed for each of the system to store 12 bits per pixel ? How much storage is required for each system if 24 bit per pixel are to be stored.

Q.N.4: Consider three different raster system with resolution 640 by 400, 1280 by 1024 and 2560 by 2048. what size frame buffer (in byte) is needed for each of the system to store 12 bits per pixel ? How much storage is required for each system if 24 bit per pixel are to be stored.

Ans :

1) for 12 bit per pixel system

the frame buffer(in byte) size needed

a. for 640 by 400 resolution = $(640 \times 400 \times 12) / 8$
=384000 bytes

b. for 1280 by 1024 resolution = $(1280 \times 1024 \times 12) / 8$
=1966080 bytes

c. for 2560 by 2048 resolution = $(2560 \times 2048 \times 12) / 8$
=7864320 bytes

2) for 24 bit per pixel system

the frame buffer(in byte) size needed

a. for 640 by 400 resolution = $(640 \times 400 \times 24) / 8$
=768000 bytes

b. for 1280 by 1024 resolution = $(1280 \times 1024 \times 24) / 8$
=3932160 bytes

c. for 2560 by 2048 resolution = $(2560 \times 2048 \times 24) / 8$
=15728640 bytes

Q.N.5. : A system with 24 bits per pixel and resolution of 1024 by 1024. Calculate the size of frame buffer (in Megabytes).

Q.N.5. : A system with 24 bits per pixel and resolution of 1024 by 1024. Calculate the size of frame buffer (in Megabytes).

Ans :

Frame size in bits= $24 \times 1024 \times 1024$ bits

Frame size in bytes= $24 \times 1024 \times 1024 / 8$ bytes

(since, 8 Bits = 1 Byte)

Frame size in kilobytes= $24 \times 1024 \times 1024 / 8 \times 1024$ KB

(since, 1024 Bytes = 1 KB)

So, Frame size in megabytes= $24 \times 1024 \times 1024 / 8 \times 1024 \times 1024$ MB

(since, 1024 KB = 1 MB)

= 3 MB.

Q.N.6. : Find out the aspect ratio of the raster system using 8 x 10 inches screen and 100 pixel/inch.



Q.N.6. : Find out the aspect ratio of the raster system using 8 x 10 inches screen and 100 pixel/inch.

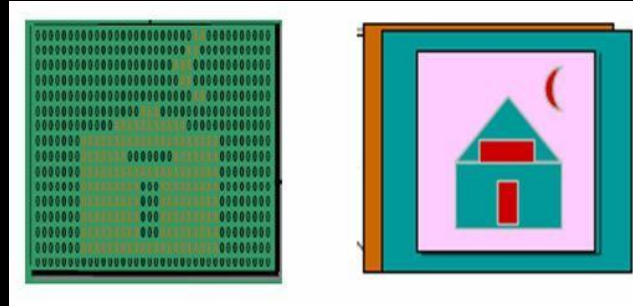
Ans :

$$\begin{aligned}\text{We know that, Aspect ratio} &= \text{Width} / \text{Height} \\ &= \frac{8 \times 100}{10 \times 100} = 4 / 5\end{aligned}$$

So, aspect ratio = 4 : 5.

Frame Buffer

- The information in the buffer typically consists of color values for every pixel .
- A frame buffer may be thought of as computer memory organized as a two-dimensional array . with each (x, y) addressable location corresponding to one pixel.
- Bit planes or bit depth is the number of bit corresponding to each pixel.



Video Display Devices (Monitors)

- A **monitor** is an output device that displays images, videos, and graphical information generated by a computer.

Functions of a Monitor

- Displays processed graphical output.
- Converts digital signals into visual images.
- Provides user interaction through visual feedback.



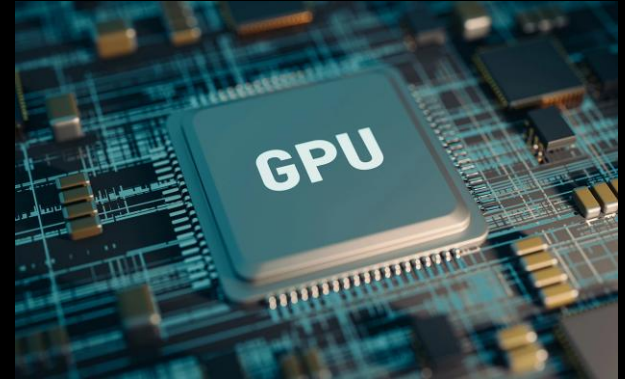
The primary output device for a graphics system is the video monitor. Historically dominated by CRTs, modern systems rely entirely on flat-panel displays.

GPU (Graphics Processing Units)

- A GPU (Graphics Processing Unit) is a specialized electronic processor designed to perform graphics calculations efficiently.

Functions of GPU

- Rendering 2D and 3D graphics
- Image processing
- Video decoding and encoding
- AI and Machine Learning computations
- Scientific simulations



GPU (Graphics Processing Units)

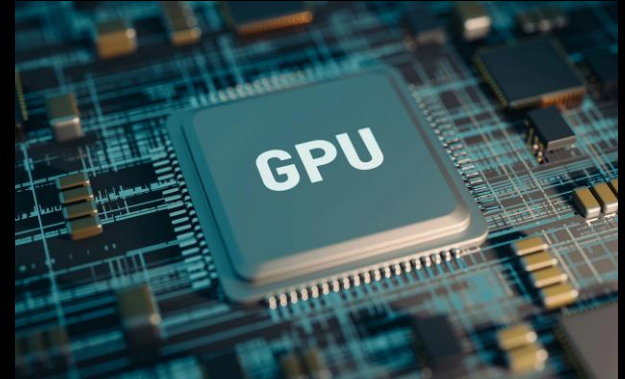
Components of a GPU

- Processing Cores
- Memory (VRAM)
- Texture Units
- Rasterizers
- Shader Units

Advantages

- High parallel processing capability
- Faster graphics rendering
- Supports complex visual effects

Examples: NVIDIA GeForce Series, AMD Radeon Series, Intel Arc Graphics

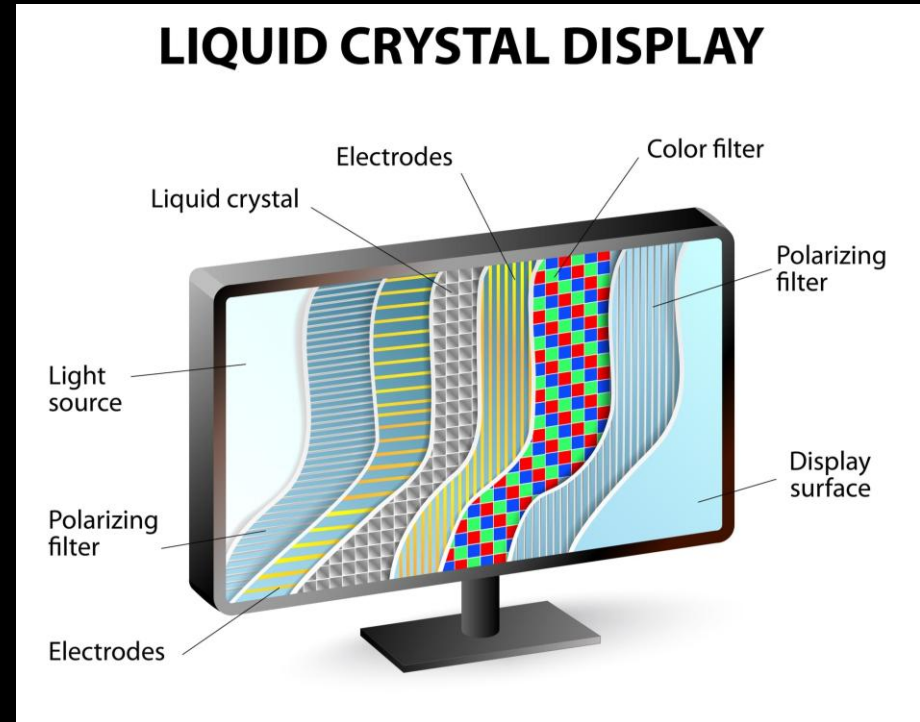


LCD (Liquid Crystal Display)

- A type of flat-panel display
- Uses liquid crystals between two sheets of material to present information on a screen

Working Principle

- Backlight generates light.
- Liquid crystals control light transmission.
- Color filters create RGB colors.
- Pixels form images on the screen.



OLED (Organic Light Emitting Diode)

OLED is a display technology where each pixel emits its own light.

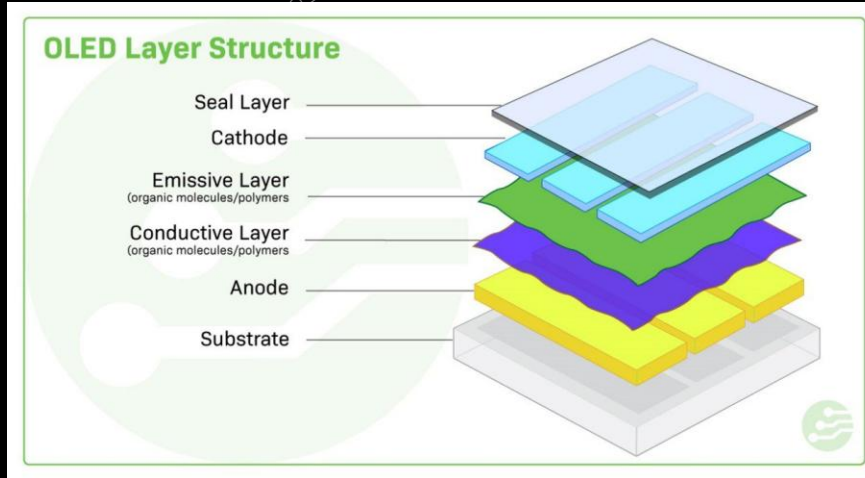
Working Principle

- Organic compounds emit light when electricity passes through them.
- No backlight is required.

Advantages over LCD:

- **True Blacks:** Since individual pixels can be turned completely off, OLEDs offer an infinite contrast ratio.
- **Faster Response Time:** Near-instantaneous pixel switching, minimizing motion blur.
- **Thinner and Flexible:** No bulky backlight unit required.

OLED (Organic Light Emitting Diode)



XP
pen

OLED vs LCD

A Comprehensive Comparison



Graphics Accelerators

A Graphics Accelerator is hardware designed to speed up graphics operations by offloading tasks from the CPU.

Functions

- Faster rendering
- Video processing
- Image enhancement
- Real-time 3D graphics

Benefits

- Improves system performance
- Reduces CPU workload
- Enables advanced visual effects



Examples

- Dedicated GPUs
- Integrated Graphics
- AI Accelerators in modern GPUs

Virtual Reality

- **Virtual reality** or **virtual realities (VR)**, also known as **immersive multimedia** or **computer-simulated reality**, is a computer technology that replicates an environment, real or imagined, and simulates a user's physical presence and environment to allow for user interaction.
- Virtual realities artificially create sensory experience, which can include sight, touch, hearing, and smell.
- Most up-to-date virtual realities are displayed either on a computer monitor or with a virtual reality headset (also called **head mounted display**).



Why Virtual Reality ? / Application of VR



- **Immersive Experience** :VR provides a highly immersive experience by creating a simulated environment that surrounds the user, making them feel present in a different reality.
- **Training and Simulation**: VR is extensively used for training purposes, allowing individuals to practice in realistic scenarios without real-world consequences. This is valuable in fields like aviation, healthcare, and military training.
- **Education and Learning**: Virtual reality enhances educational experiences by allowing students to explore historical sites, conduct virtual dissections, or engage in interactive learning environments.
- **Entertainment and Gaming**: VR enhances gaming experiences by providing players with a more immersive and interactive environment. It creates a sense of presence and engagement.
- **Virtual Travel**: VR allows users to virtually explore different locations and experiences without leaving their homes, making it a valuable tool for tourism and travel-related industries.

Why Virtual Reality ? / Application of VR

- **Architectural Visualization:** In architecture and design, VR enables professionals to visualize and explore 3D models of buildings and spaces before construction, facilitating better planning and decision-making.
- **Data Visualization:** Complex data sets can be visualized in 3D environments, making it easier for researchers and analysts to explore and understand intricate patterns and relationships.
- **Remote Collaboration:** VR enables teams to collaborate in virtual environments, making it possible for remote workers to feel more connected and engaged during meetings and collaborative projects.
- **Mental Health and Relaxation:** VR applications offer relaxation experiences, meditation environments, and stress-relief scenarios, contributing to mental health and well-being.



Augmented Reality (AR)



Augmented Reality (AR)

- **Augmented Reality (AR)** is a perfect blend of the digital world and the physical elements to create an artificial environment.
- Apps which are developed using AR technology for mobile or desktop to blend digital components into the real world. The full form of AR is Augment Reality.
- Example: AR technology helps to display score overlays on telecasted sports games and pop out 3D photos, text messages, and emails.



Augmented Reality (AR) Vs Virtual Reality (VR)

AR	VR
The system augments the real-world scene	Completely immersive virtual environment
In AR User always have a sense of presence in the real world	In VR, visual senses are under control of the system
AR is 25% virtual and 75% real	VR is 75% virtual and 25% real
This technology partially immerses the user into the action	This technology fully immerses the user into the action
No AR headset is needed.	Some VR headset device is needed.
With AR, end-users are still in touch with the real world while interacting with virtual objects nearer to them.	By using VR technology, VR user is isolated from the real world and immerses himself in a completely fictional world.
It is used to enhance both real and virtual worlds.	It is used to enhance fictional reality for the gaming world.

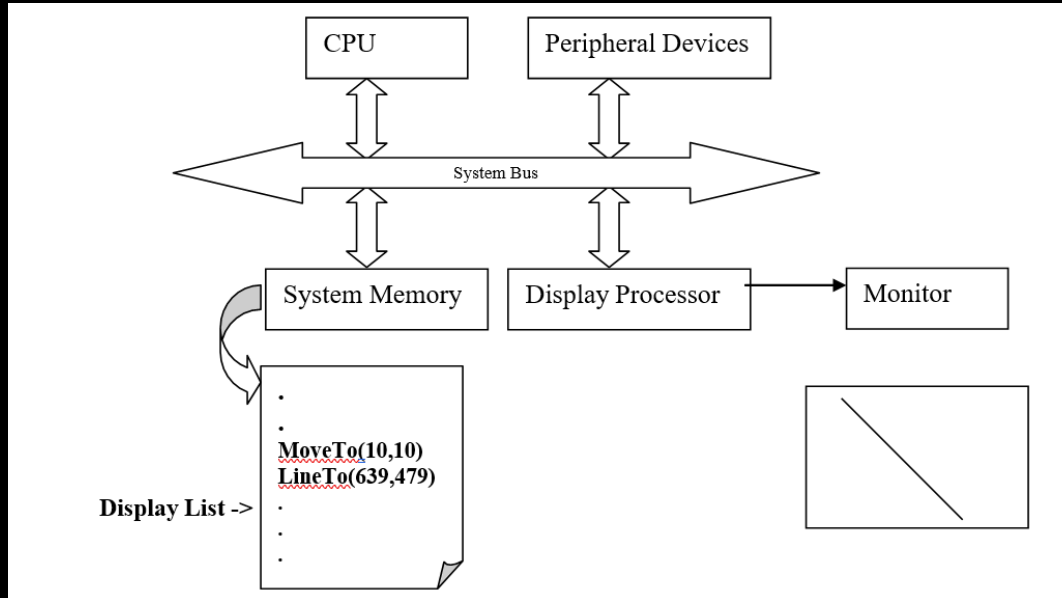
Raster and Random Scan Displays and Architecture, Display Processor, Graphics Pipeline: From Modeling to Rendering, Brief History: From Raster/CRT to Real-Time Rendering, Input Devices

Raster Graphics Vs Vector Graphics: Digital graphics types

Raster Graphics	Vector Graphics
Made up of pixels or dots	Made up of mathematical equations, lines and curves
Resolution dependent	Resolution independent
Image quality deteriorates	Image quality remains constant
Scaling results in loss of data	Scaling does not affect the quality
Limited to rectangular shapes	Can create any shape
Larger file size for high-res	Smaller file size for high-res
Suitable for photographs and images	Suitable for logos and designs
Editing is often difficult	Editing is easy and flexible
Examples: JPEG, PNG, gif, bmp, tiff, psd	Examples: SVG, AI, eps, dae, ps, emf

Vector Technology

- Vector display technology was developed in 60"s and used as a common display device until 80"s , It is also called *random scan*, *a stroke*, *a line drawing* or *calligraphic display*.



Vector Technology

- It consists of **a central processing unit, a display processor , a monitor , system memory** and peripheral devices such as mouse and key board
- A **display processor** is also called a *display processing unit* or *graphics controller*
- The application program and *graphics subroutine package* both reside in the system memory and execute on CPU .
A graphics subroutine package creates a *display list* and stores in the system memory
- A display list contains point and line plotting commands with end point coordinates as well as character plotting commands
- The DPU interprets the commands in the display list and plots the respective output primitives such as point, line and characters
- As a matter of fact the DPU sends digital point coordinates to a vector generator that converts the digital coordinate values to analog voltages for circuits that displace an electron beam hitting on the CRT's phosphor coating.

Vector Technology

- Therefore, *the beam is deflected from endpoint to endpoint as dictated by the arbitrary order of the commands in the display list*, hence the name *Random Scan Display*
- Since the light output of the phosphor decays in tens or at most hundreds of microseconds the DPU must cycle thru the display list to refresh the image around 50 times per second to avoid flicker. A portion of the system memory where display list resides is called a ***refresh buffer***.
- This display technology is used with mono chromatic CRTs or beam penetration color CRTs.

Vector Technology

Advantages:

- It can **produce a smooth output primitives with higher resolution** unlike the raster display technology
- It is **better than raster display for real time dynamics such as animation**
- For transformation, only the end points has to be moved to the new position in vector display but in raster display it is necessary to move those end points and at the same time all the pixels between the end points must be scan converted using appropriate algorithm No prior information on pixels can be reused

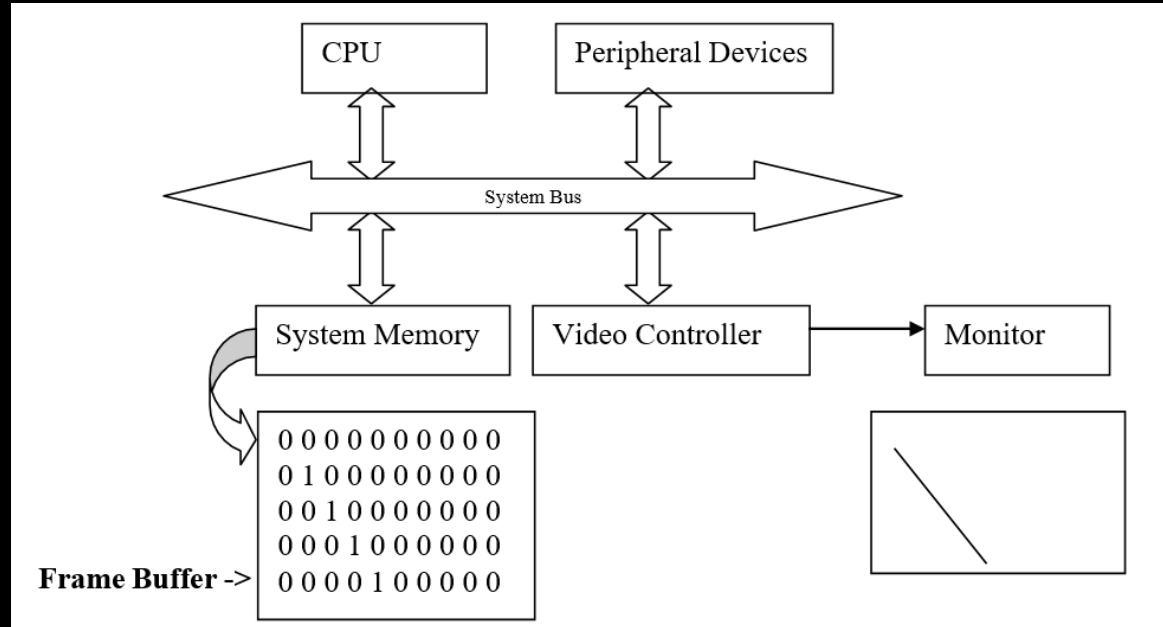
Disadvantages:

- A vector display **can not fill areas with patterns and manipulative bits**
- Time required **for refreshing an image depends upon its complexity** (more the lines , longer the time) the flicker may therefore appear as the complexity of the image increases. The fastest vector display can draw about 100000 short vectors in a refresh cycle without flickering

Raster Display Technology

This technology based on television technology was developed in early 70s

- It consists of central processing unit, a video controller, a monitor, system memory and peripheral devices such as mouse and keyboard.
- The application program and graphics subroutine package both reside in the system memory and execute on CPU.



Raster Display Technology

- When a particular command such as a $\text{line}(x1, y1, x2, y2)$ is called by the application program the graphics subroutine package sets the appropriate pixels in the *frame buffer*, a portion of the system memory
- The *video controller* then cycles thru the frame buffer, one scan line at a time typically 50 times per second.
- It brings a value of each pixel contained in the buffer and uses it to control the intensity of the CRT electron beam.

Raster Display Technology

Advantages

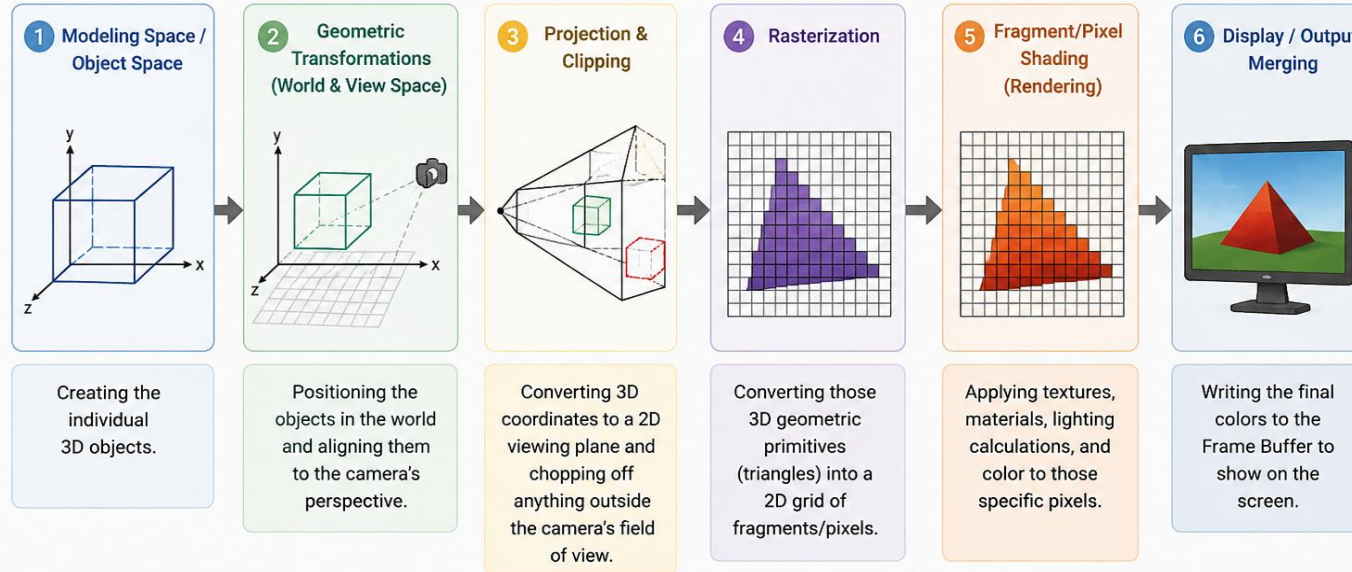
- It has an ability to fill the areas with solid colors or patterns
- The time required for refreshing is independent of the complexity of the image
- Low cost

Disadvantages

- For Real-Time dynamics not only the end points are required to move but all the pixels in between the moved end points have to be scan converted with appropriate algorithms Which might slow down the dynamic process
- Due to scan conversion “jaggies” or “stair-casing” are unavoidable

Graphics Pipeline: From Modeling to Rendering

Graphics Pipeline: From Modeling to Rendering



Graphics Pipeline: From Modeling to Rendering

1. **Modeling Space / Object Space:** Creating the individual 3D objects.
2. **Geometric Transformations (World & View Space):** Positioning the objects in the world and aligning them to the camera's perspective.
3. **Projection & Clipping:** Converting 3D coordinates to a 2D viewing plane and chopping off anything outside the camera's field of view.
4. **Rasterization:** Converting those 3D geometric primitives (triangles) into a 2D grid of fragments/pixels.
5. **Fragment/Pixel Shading (Rendering):** Applying textures, materials, lighting calculations, and color to those specific pixels.
6. **Display / Output Merging:** Writing the final colors to the Frame Buffer to show on the screen.

Cathode Ray Tube (CRT)

Cathode Ray Tube (CRT)

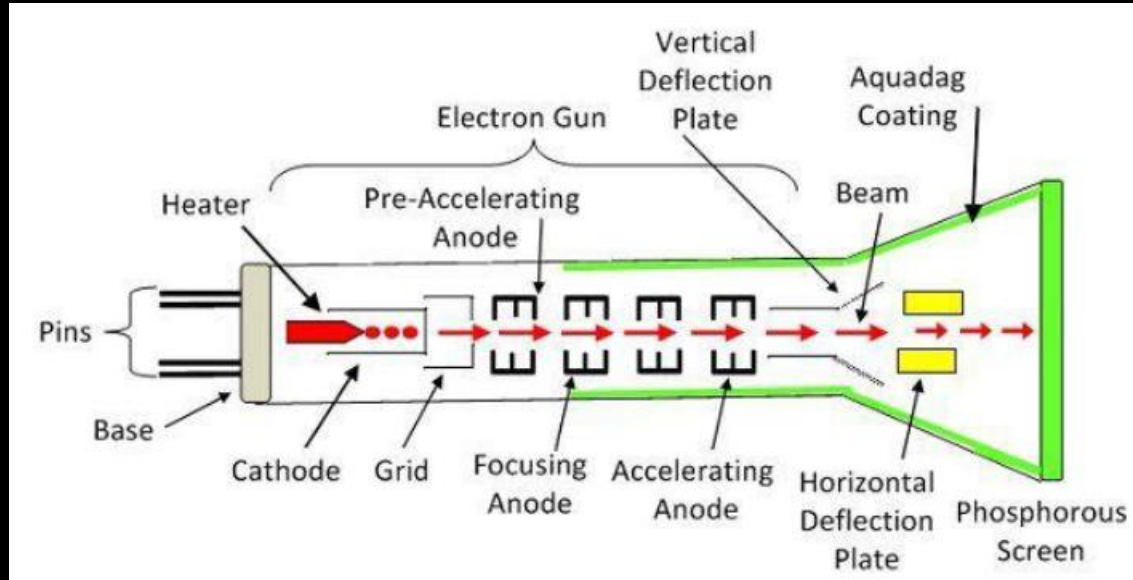


Figure 3: CRT

Video Display Devices

■ Cathode Ray Tube (CRT)

- Components:
 - Electron Gun: composed of heated metal cathode and control grid
 - Accelerating Anode
 - Focusing System
 - Deflection system
 - Phosphor Screen

Cathode Ray Tube (CRT): Steps

It operates based on the principles of electron emission and manipulation of electron beams to create images on a phosphor-coated screen. The working contains these steps:

- 1. Electron Emission :** The cathode (heated filament) emits electrons. These electrons are directed into a beam by the electron gun.
- 2. Electron Acceleration:** The emitted electrons are accelerated by high voltage electrodes, creating a high speed electron beam.
- 3. Beam focusing:** Focusing coils or electrodes shape the electron beam into a fine, narrow stream to ensure precision when hitting the screen.
- 4. Beam Deflection:** Magnetic or electrostatic deflection plates steer the electron beam horizontally and vertically across the screen to trace images line by line.
- 5. Phosphor Screen Illumination:** The electron beam strikes the phosphor coated screen causing it to emit light at the points of impact, forming the image.

Graphics Software: Software standards, Need of machine independent graphics language

Software Standards

Graphics software refers to computer programs that allow users to create, edit, and manipulate visual elements such as images, illustrations, and designs. Some of the common categories of graphics software are:

1. **Raster Graphics Editors:** Adobe Photoshop, GIMP (GNU Image Manipulation Program), etc.
2. **Vector Graphics Editors:** Adobe Illustrator, Inkscape, etc.
3. **Other software** like Adobe InDesign, Autodesk Maya, Blender, Lightroom, Adobe Premier Pro etc.

Software Standards in Graphics

- **Graphics software** standards play a crucial role in ensuring interoperability, consistency, and portability across different graphics applications and systems.
- These standards help define common protocols, interfaces, and procedures that facilitate communication between software components and hardware. Here are some notable graphics software standards:
 1. Graphics Kernel System (GKS)
 2. Programmer's Hierarchical Interactive Graphics System (PHIGS)
 3. Computer Graphics Metafile (CGM)

Software Standards in Graphics

Graphics Kernel System (GKS):

- ✓ **Overview:** GKS is an ISO standard (ISO 7942) that defines a common interface for 2D vector graphics applications. It provides a set of functions for drawing and manipulating graphical primitives such as lines, circles, and polygons.
- ✓ **Purpose:** GKS aims to promote portability of graphics applications across different hardware and software platforms by providing a standardized interface.
- ✓ **Features:** Supports both basic and high-level graphics operations. Defines a device-independent logical input and output model.

Software Standards in Graphics

Programmer's Hierarchical Interactive Graphics System (PHIGS)

- ✓ **Overview:** PHIGS is an ISO standard (ISO/IEC 9592) that builds upon GKS and extends its capabilities to include three-dimensional graphics and hierarchical modeling.
- ✓ **Purpose:** PHIGS provides a comprehensive set of functions for interactive 3D graphics programming, including features for transformations, modeling, and rendering.
- ✓ **Features:** Supports hierarchical modeling, input devices, and the specification of attributes for graphical entities. PHIGS-PLUS and PEX (PHIGS Extension to X) are extensions of PHIGS.

Software Standards in Graphics

Computer Graphics Metafile (CGM)

- ✓ **Overview:** CGM is an international standard (ISO/IEC 8632) that defines a metafile format for the interchange of 2D vector and mixed vector/raster graphics.
- ✓ **Purpose:** CGM facilitates the exchange of graphics data between different systems and applications by providing a standard representation for graphical elements.
- ✓ **Features:** Supports a wide range of graphical elements, including lines, text, polygons, and raster images. Allows for the inclusion of descriptive information.



Need of machine independent graphics language

A machine-independent graphics language is essential for several reasons, as it addresses challenges associated with the diversity of computer hardware and software platforms. Here are key reasons for the need of such a language:

1. Portability
2. Interoperability
3. Ease of Development
4. Standardization
5. Reduced Maintenance Overhead
6. Cost Savings
7. Scalability
8. Accessibility
9. Global Collaboration

Questions for Assignment 1

1. Define Computer Graphics. Explain any five applications of Computer Graphics in different fields.
2. Discuss the evolution of Computer Graphics from CRT displays to modern real-time rendering systems.
3. What is the Graphics Pipeline? Explain the different stages involved from Modeling to Rendering.
4. Compare CRT, LCD and OLED display technologies based on working principle, advantages, and disadvantages.
5. What is Graphics Software? Explain different categories of graphics software with examples.
6. Differentiate between Raster Scan Display and Random Scan Display.
7. Explain the need for machine-independent graphics languages. How do such standards benefit application developers?
8. What do you mean by the refresh rate of a display device? Explain vector scan display with a suitable diagram.
9. Differentiate between virtual reality and augmented reality with examples.
10. Differentiate between raster and vector graphics method.
11. Calculate the total memory required to store a 5-minute video in SVGA system with 24-bit true color and 30 fps.
12. Explain software standards in computer graphics.
13. Define pixel, resolution, refresh rate and aspect ratio in brief.

Compiled by ab

THANK YOU
Any Queries ?